

This **non- equi join** produces all combinations where an employee's department is **not equal** to the department row's DEPT. Let's trace it step by step with the EMP and DEPART tables from the slide.

Step 1: Original tables (same as equi- join example)

EMP (4 rows):

NAME	OFFICE	DEPT	SALARY
Sumith	400	CS	25000
Ravi	220	Econ	30000
Kiran	160	Elec	35000
Sathish	420	Econ	40000

DEPART (4 rows):

DEPT	MAIN OFFICE	PHONE
CS	404	2436789
Econ	200	2378658
Fin	501	2356297
Hist	100	2745897

Step 2: Cartesian product EMP × DEPART

16 rows total (4 × 4 combinations). Every employee is paired with every department.

For example, the first 4 rows would be:

NAME	OFFICE	EMP.DEPT	SALARY	DEPART.DEPT	MAIN OFFICE	PHONE
Sumith	400	CS	25000	CS	404	2436789
Sumith	400	CS	25000	Econ	200	2378658
Sumith	400	CS	25000	Fin	501	2356297
Sumith	400	CS	25000	Hist	100	2745897
...	...	...	...	...	...	...

Step 3: Selection $\sigma_{\{EMP.DEPT \neq DEPART.DEPT\}}$

Apply the non- equi condition: keep only rows where $EMP.DEPT \neq DEPART.DEPT$.

For Sumith (CS): Drop the row where $DEPART.DEPT = CS$; **keep** the other 3: Econ, Fin, Hist.

For Ravi (Econ): Drop Econ; **keep** CS, Fin, Hist.

For Kiran (Elec): Elec does not exist in DEPART, so **all 4** rows are kept (CS, Econ, Fin, Hist).

For Sathish (Econ): Drop Econ; **keep** CS, Fin, Hist.

Total kept rows: $3 + 3 + 4 + 3 = 13$ rows.

Step 4: Complete final result table

The **non- equi join result** has **13 rows** (all columns from both tables):

Sumith (CS) $\times$ {Econ, Fin, Hist}

NAME	OFFICE	EMP.DEPT	SALARY	DEPART.DEPT	MAIN OFFICE	PHONE
Sumith	400	CS	25000	Econ	200	2378658
Sumith	400	CS	25000	Fin	501	2356297
Sumith	400	CS	25000	Hist	100	2745897

Ravi (Econ) × {CS, Fin, Hist}

NAM E	OFFIC E	EMP.DEP T	SALAR Y	DEPART.DEP T	MAIN OFFICE	PHONE
Rav i	220	Econ	30000	CS	404	2436789
Rav i	220	Econ	30000	Fin	501	2356297
Rav i	220	Econ	30000	Hist	100	2745897

Kiran (Elec) × {CS, Econ, Fin, Hist}

NAME	OFFIC E	EMP.DEP T	SALAR Y	DEPART.DEP T	MAIN OFFICE	PHONE
Kira n	160	Elec	35000	CS	404	2436789
Kira n	160	Elec	35000	Econ	200	2378658
Kira n	160	Elec	35000	Fin	501	2356297
Kira n	160	Elec	35000	Hist	100	2745897

Sathish (Econ) × {CS, Fin, Hist}

NAME	OFFIC E	EMP.DEP T	SALAR Y	DEPART.DEP T	MAIN OFFICE	PHONE
Sathis h	420	Econ	40000	CS	404	2436789
Sathis h	420	Econ	40000	Fin	501	2356297
Sathis h	420	Econ	40000	Hist	100	2745897

Notice:

- **Only 3 rows dropped:** the “matching” ones (Sumith- CS, Ravi- Econ, Sathish- Econ).
- **DEPT appears twice** (emp.DEPT and depart.DEPT), just like equi- join.
- **Kiran gets 4 rows** because his Elec department has no match in DEPART, so all combinations satisfy “≠”.
- The result is **large** (13 out of 16) because most combinations naturally have different departments.

Step 5: Intuition and use cases

What question does this answer?

“List every employee together with every department **except** the one they belong to.”

Real- world use cases for non- equi joins:

- Salary ranges: $EMP \bowtie _ \{EMP.Salary \text{ BETWEEN } Dept.MinSalary \text{ AND } Dept.MaxSalary\} Dept$
- Dates: $EMP \bowtie _ \{EMP.JoinDate > Dept.FoundedYear\} Dept$
- Geographic ranges: $EMP \bowtie _ \{EMP.ZipCode \text{ BETWEEN } Dept.ZipStart \text{ AND } Dept.ZipEnd\} Dept$

Key difference from equi- join:

- Equi- join ($DEPT = DEPT$): would return only **3 rows** (Sumith- CS, Ravi- Econ, Sathish- Econ).
- Non- equi ($DEPT \neq DEPT$): returns **13 rows** (everything else).

The slide shows only the first few rows to illustrate the pattern; the full table is all 13 combinations above.